

Matrices

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Learning Outcomes

At the end of this lecture, you should be able to:

- ☐ Know what is a matrix and its structure.
- ☐ Recognize some special matrices.
- ☐ Performs operations involving matrices.
- ☐ Knows some properties on matrix addition and matrix multiplication.
- ☐ Know what is the transpose of a matrix and its properties.
- ☐ Know what is a skew matrix and what is a skew-symmetric matrix.

Introduction to Matrices

Let $\mathbb{F}$ be a real field or complex field. An $m \times n$ (m by n) matrix over $\mathbb{F}$ is an array of numbers arranged in m rows and n columns.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix} = [a_{ij}]_{m \times n}$$

The a_{ij} is called (i, j) -entry of A , which is the number in the i^{th} row and j^{th} column of the matrix A .

A set of all $m \times n$ matrices over the field $\mathbb{F}$ is denoted by $M_{m,n}(\mathbb{F})$ or simply $M_{m,n}$.

We always refer to the rows before the columns when describing the size of a matrix or the position of an entry.

Introduction to Matrices

A **column matrix** (or **column vector**) is a matrix with only one column, that is, it is an $m \times 1$ matrix.

A **row matrix** (or **row vector**) is a matrix with only one row, that is, it is an $1 \times n$ matrix.

A matrix is called a **square matrix** if it has the same number of rows and columns ($m = n$).

An $n \times n$ matrix is called a **square matrix of order n** or **n -square matrix**, and denoted by $M_n(\mathbb{F})$.

The **diagonal** (or **main diagonal**) of a n -square matrix A is the sequence of entries $a_{11}, a_{22}, \dots, a_{nn}$.

The entries a_{ii} are called the **diagonal entries** of A , while a_{ij} , $i \neq j$, are called the **off-diagonal** or **non-diagonal entries** of A .

Introduction to Matrices

Example:

$$\begin{bmatrix} -2 & 1 & -1 & 0 \\ 0 & 3 & 1 & 2 \end{bmatrix} \in M_{2,4}(\mathbb{R}) \text{ is a } 2 \times 4 \text{ matrix.}$$

$$[3 \quad 1+i \quad i \quad -5 \quad 7] \in M_{1,5}(\mathbb{C}) \text{ is a } 1 \times 5 \text{ row matrix.}$$

$$\begin{bmatrix} 1 \\ 3 \\ 0 \\ -4 \end{bmatrix} \in M_{4,1}(\mathbb{R}) \text{ is a } 4 \times 1 \text{ column matrix.}$$

$$\begin{bmatrix} 1 & 2 & 4 \\ -1 & 0 & 3 \\ 3 & 1 & -2 \end{bmatrix} \in M_3(\mathbb{R}) \text{ is a 3-square matrix.}$$

Some Special Square Matrices

A **diagonal matrix** is a square matrix $A = [a_{ij}]$ whose non-diagonal entries are all zero ($a_{ij} = 0 \forall i \neq j$) and is denoted by $A = \text{diag}(a_{11}, a_{22}, \dots, a_{nn})$.

An **identity matrix** (or **unit matrix**) is a diagonal matrix with all its diagonal entries are 1 ($a_{ii} = 1 \forall i = 1, 2, \dots, n$ and $a_{ij} = 0 \forall i \neq j$) and is denoted by I_n .

A **scalar matrix** is a diagonal matrix with all its diagonal entries are the same. ($a_{ii} = c \forall i = 1, 2, \dots, n$ and $a_{ij} = 0 \forall i \neq j$, for some constant c). A scalar matrix can be written as cI_n .

A **zero matrix** is an $m \times n$ matrix with all entries equal to zero ($a_{ij} = 0 \forall i = 1, 2, \dots, m$ and $j = 1, 2, \dots, n$, for some constant c) and is denoted by $0_{m \times n}$.

Some Special Square Matrices

An **upper triangular matrix** is a square matrix $A = [a_{ij}]$ whose entries below the main diagonal are all zero ($a_{ij} = 0 \ \forall i > j$, for some constant c).

An **lower triangular matrix** is a square matrix $A = [a_{ij}]$ whose entries above the main diagonal are all zero ($a_{ij} = 0 \ \forall i < j$, for some constant c).

Both upper and lower triangular matrices are called triangular matrices.

A **symmetric matrix** is a square matrix $A = [a_{ij}]$ whose transpose equals the matrix ($A = A^T$). That is, $a_{ij} = a_{ji} \ \forall i, j$.

A **skew-symmetric matrix** is a square matrix $A = [a_{ij}]$ whose transpose equals the negative of the matrix ($A = -A^T$). That is, $a_{ij} = -a_{ji} \ \forall i, j$.

Some Special Square Matrices

Example:

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 0 & -8 \\ 0 & -2 & 5 \\ 0 & 0 & -7 \end{bmatrix}, C = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 6 & 1 & 0 & 0 \\ 5 & -2 & -8 & 0 \\ 0 & 7 & 0 & 4 \end{bmatrix}, D = \begin{bmatrix} 1 & 3 & -1 \\ 3 & -2 & 0 \\ -1 & 0 & 7 \end{bmatrix},$$

$$E = \begin{bmatrix} -1 & 6 & 1 & 0 \\ -6 & 0 & 2 & 7 \\ -1 & -2 & -8 & -3 \\ 0 & -7 & 3 & 4 \end{bmatrix}.$$

Matrix Operations

Let $A = [a_{ij}]_{m \times n}$, $B = [b_{ij}]_{m \times n}$, and $C = [c_{ij}]_{m \times n}$ be three $m \times n$ matrices.

- **Equality:** $A = B \leftrightarrow a_{ij} = b_{ij}$ for all i and j .
- **Matrix Addition:** $A + B = C \leftrightarrow c_{ij} = a_{ij} + b_{ij}$ for all i and j .
- **Matrix Subtraction:** $A - B = C \leftrightarrow c_{ij} = a_{ij} - b_{ij}$ for all i and j .
- **Scalar Multiplication:** $\lambda A = C \leftrightarrow c_{ij} = \lambda a_{ij}$ for all i and j , for any scalar $\lambda \in \mathbb{F}$.

Given a matrix A , $-A$ is used to denote $(-1)A$.

Given two matrices A and B of the same size, $A - B$ is defined to be $A + (-B)$.

Matrix Operations

Matrix Multiplication: If $A = [a_{ij}]_{m \times r}$ is an $m \times r$ matrix and $B = [b_{ij}]_{r \times n}$ is a $r \times n$ matrix, then the matrix product AB is the $m \times n$ matrix whose (i, j) -entry is given by

$$a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ir}b_{rj} = \sum_{k=1}^r a_{ik}b_{kj}, i = 1, \dots, m, j = 1, \dots, n.$$

Matrix Operations

Example:

$$\text{Let } A = \begin{bmatrix} -2 & 1 & -1 & 0 \\ 0 & 3 & 1 & 2 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 & 2 & 1 \\ 4 & 1 & 6 & 2 \end{bmatrix}, C = \begin{bmatrix} -1 & 8 \\ 2 & 1 \\ 1 & 1 \\ 12 & 6 \end{bmatrix}.$$

$$A + B =$$

$$A - B =$$

Matrix Operations

Example:

$$\text{Let } A = \begin{bmatrix} -2 & 1 & -1 & 0 \\ 0 & 3 & 1 & 2 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 & 2 & 1 \\ 4 & 1 & 6 & 2 \end{bmatrix}, C = \begin{bmatrix} -1 & 8 \\ 2 & 1 \\ 1 & 1 \\ 12 & 6 \end{bmatrix}.$$

$$BC =$$

$$CB =$$

Matrix Operations

Example:

$$\text{Let } A = \begin{bmatrix} -2 & 1 & -1 & 0 \\ 0 & 3 & 1 & 2 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 & 2 & 1 \\ 4 & 1 & 6 & 2 \end{bmatrix}, C = \begin{bmatrix} -1 & 8 \\ 2 & 1 \\ 1 & 1 \\ 12 & 6 \end{bmatrix}.$$

$$-3C =$$

Properties of the Matrix Addition

Let A, B, C be three $m \times n$ matrices. Then

- **Commutative Law:** $A + B = B + A$.
- **Associative Law:** $(A + B) + C = A + (B + C)$.
- **Distributive Law:** $\lambda(A + B) = \lambda A + \lambda B$ and $(\lambda + \beta)A = \lambda A + \beta A$, for any constants λ and β .

Properties of the Matrix Multiplication

Let $A, A_1, A_2 \in M_{m \times n}(\mathbb{F})$, $B, B_1, B_2 \in M_{n \times p}(\mathbb{F})$ and $C \in M_{p \times q}(\mathbb{F})$. Then

- **Associative Law:** $(AB)C = A(BC)$.
- **Distributive Law:** $\lambda(AB) = (\lambda A)B = A(\lambda B)$ for any constant λ ,
 $A(B_1 + B_2) = AB_1 + AB_2$,
 $(A_1 + A_2)B = A_1B + A_2B$.
- **Identity Law:** $AI_n = A = I_nA$.

In general, $AB \neq BA$.

We called AB as the **pre-multiplication** of A to B .

We called BA as the **post-multiplication** of A to B

$AB = 0$ does not imply that $A = 0$ or $B = 0$ or $BA = 0$.

Exercise: Can we apply binomial theorem to the matrices? Is it true that $(A + B)^2 = A^2 + 2AB + B^2$?

Transpose

Let $A \in M_{m \times n}(\mathbb{F})$ be a $m \times n$ matrix. The **transpose** of the matrix A , denoted by A^T , is a $n \times m$ matrix obtained from A , by changing its columns to rows (and rows to columns). That is

$$[A^T]_{ij} = [A]_{ji}.$$

Example:

$$A = \begin{bmatrix} 1 & 0 & -8 \\ 3 & -2 & 5 \\ 6 & 9 & -7 \end{bmatrix}$$

$$C = \begin{bmatrix} 3 & 1 & 2 \\ 4 & 6 & 5 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 3 \\ 1 & 2 \\ 4 & 5 \end{bmatrix}$$

Properties of Transpose of Matrix

Let $A = [a_{ij}]$ be an $m \times n$ matrix. Then

- $(A^T)^T = A$.
- $(A + B)^T = A^T + B^T$ if $B = [b_{ij}]_{m \times n}$.
- $(\lambda A)^T = \lambda A^T$ for some constant λ .
- $(AB)^T = B^T A^T$ if $B = [b_{ij}]_{n \times p}$

Symmetric and skew-symmetric matrix

An $n \times n$ matrix A is said to be **symmetric** if and only if $A^T = A$.

And, an $n \times n$ matrix A is said to be **skew-symmetric** if and only if $A^T = -A$.

Example: $B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 6 & 4 \\ 3 & 4 & 5 \end{bmatrix}$ is symmetric and $C = \begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & 3 \\ -2 & -3 & 0 \end{bmatrix}$ is skew-symmetric.

Note: A symmetric and a skew-symmetric matrix must necessarily a square matrix.

Symmetric and skew-symmetric matrix

Example:

Let $A = \begin{bmatrix} 3 & -5 & a \\ b & 8 & c \\ 0 & d & e \end{bmatrix}$. Determine the values of a, b, c, d, e so that A is symmetric.

Show that $C = B^T B$ is symmetric for any matrix B .